DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
West Point, New York 10996

MADN-MTH 2 November 2026

MEMORANDUM FOR DXDT "Velocity" Scholarship Committee, United States Military Academy, West Point, NY 10996

SUBJECT: Application for AY 2027 DXDT "Velocity" Scholarship

1. Long-term Professional Goals. I intend to serve as an Army officer who turns operational problems into rigorous models and fielded software. I plan to commission as an Infantry officer and lead Soldiers first, because I believe many crucial problems start within formations. As an Operations Research major with a minor in artificial intelligence and a finance specialization, I then want to build toward service as an Operations Research/Systems Analyst (FA49) or work with AI through the Army AI Integration Center (AI2C) or the GenAI Task Force. Additionally, I plan to earn a graduate degree, such as an MBA or another qualifying degree, and return to West Point to teach in the Department of Mathematical Sciences, bringing current industry practice into the classroom.
2. Purpose. Mr. Philippone has described data as the ammunition of the future and has framed a commander's hardest choices as constrained optimization, which means finding the best decision across competing tradeoffs fast enough to matter. This past summer I saw that idea from three angles. On the Sea to Shining Sea AIAD, Palantir's team in Palo Alto walked us through Project Maven and how it gives commanders at every echelon a common picture to decide from. At the 4th Infantry Division, we saw how a formation employs those systems and feeds lessons back to the developers. At Stanford's Gordian Knot Center, Col. (Ret.) Joseph Felter and Jack Faherty described how student startups get new technology into the hands of Army units, and why that is so hard.

At my second AIAD, with the Army's Transformation Decision Analysis Center (TDAC), I built a reinforcement-learning model for contested resupply. I am pursuing the DXDT "Velocity" Scholarship for two reasons. First, its funding would take my research out of the classroom to defense conferences where I can present it, a hackathon where I can compete, and an AIAD where I can learn how tools like it are tested before they reach a commander. Second, as a Velocity Scholar I would seek to work alongside industry partners such as Palantir, whether through an AIAD, a hackathon, or a joint research project, and bring that practice back to my research and my future formation.

3. Near-term Research Goals. My near-term goal is to complete and present my research on a transportation and logistics problem that combines reinforcement learning and network optimization on a maritime network. At TDAC I redesigned the reward function for a reinforcement-learning agent that allocates resupply in Logistics Battle Command, the Army's sustainment simulation. I built diagnostic tools that exposed why a high-scoring agent was starving the priority unit, and the retrained agent more than tripled the share of requests filled.

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This semester I am extending that work to network interdiction with my advisor, LTC Matthew Webb, whose doctoral research was in transportation and logistics. My project asks where an adversary with a fixed effort budget should act to cut Taiwan’s crude oil supply. Additionally, it tests whether LTC Webb’s reinforcement-learning attacker, trained on Ukraine’s rail network, still finds strong strategies on sea lanes. I built a 57-node, 79-edge model of Taiwan’s crude import routes from open sources, with sourced capacities and seven costed adversary actions, and verified it against an independent max-flow calculation.

Over the coming year I will (1) compare optimization baselines against the trained agent, (2) run wargaming scenarios that vary blockade and cyber costs to see how the adversary’s best plan changes, and (3) report whether a rail-trained attacker transfers to a maritime network. I will present the results at Projects Day and at the conferences in paragraph 4.

4. Funding Plan and Budget.

a. Funding Plan. My plan is one AIAD, three conferences in priority order, and three online courses. I request funding for the AIAD, my first-priority conference, and the courses; the second and third conferences are alternates if a higher priority falls through.

   AIAD. From 11–31 July 2027, I will spend $6,700 on housing, food, and transportation for a three-week AIAD at MIT Lincoln Laboratory in Lexington, MA, a Department of Defense research center that builds and tests AI and decision-support tools for the military. At TDAC, my agent scored well while starving the priority unit, which is a testing and evaluation failure. At Lincoln Laboratory I will learn how engineers test these tools before they reach Soldiers. If a Lincoln Laboratory placement is not available, my alternate is a research AIAD at Duke University’s Department of Electrical and Computer Engineering, where LTC Patrick Kuiper earned his doctorate, at the same estimated cost.

   Conference, Priority 1. On 24–25 May 2027, I will spend $2,000 on train travel, lodging, and meals to attend the SCSP AI+ Expo in Washington, DC; admission is free. The Expo is the largest gathering of defense-technology companies and government AI leaders in the country, with more than 18,000 attendees. If SCSP runs its National Security Technology Hackathon again, I will enter with a team of cadets.

   Conference, Priority 2. In early May 2027, I will spend $2,000 on travel, lodging, meals, and registration to present my Taiwan interdiction paper at the IEEE Systems and Information Engineering Design Symposium (SIEDS) at the University of Virginia, if accepted. Accepted papers are published in IEEE Xplore, which would give the project a peer-reviewed, citable result for the department.
   Conference, Priority 3. From 16–18 March 2027, I will spend $2,000 on travel, lodging, and meals to attend the AUSA Global Force Symposium in Huntsville, AL; registration is free for military attendees. It is the Army’s annual forum on modernization, where senior leaders and industry discuss the technology, acquisition, and concepts that will shape the future force. Attending would show me which problems the Army is trying to solve and where research like mine can help.

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Online Courses. During 2027, I will spend $797 on verified certificates for three courses in the MITx Supply Chain Management MicroMasters program, offered by the MIT Center for Transportation & Logistics. The courses are Supply Chain Analytics (SC0x, $199), Supply Chain Fundamentals (SC1x, $299), and Supply Chain Design (SC2x, $299). Supply Chain Design covers network design and optimization, the defender’s side of the problem my Taiwan model attacks. The courses also count toward the MicroMasters credential, a pathway to graduate study in supply chain management.

b. Research Travel. Conferences are listed in priority order; Priorities 2 and 3 are alternates.

Purpose of Trip (AIAD, Conference, other)	**Destination**	**City, State**	**Dates or # of Days**	**Estimated total Trip costs**
AIAD	MIT Lincoln Laboratory (alternate, Duke University ECE)	Lexington, MA (alternate, Durham, NC)	11–31 Jul 2027	$6,700
Conference, Priority 1	SCSP AI+ Expo and hackathon	Washington, DC	24–25 May 2027	$2,000
Conference, Priority 2 (alternate)	IEEE SIEDS, University of Virginia	Charlottesville, VA	Early May 2027	$2,000
Conference, Priority 3 (alternate)	AUSA Global Force Symposium	Huntsville, AL	16–18 Mar 2027	$2,000

c. Equipment, Materials, or Supplies. Three MITx course certificates purchased from MIT (learn.mit.edu), SC0x ($199), SC1x ($299), and SC2x ($299), for a total of $797.

d. Total. $9,497, covering the AIAD ($6,700), the first-priority conference ($2,000), and the online courses ($797). Each alternate conference is also estimated at $2,000, so any substitution keeps the total at $9,497, within the $10,000 limit.

e. Timeline for Expenditures. From November 2026 to February 2027, I will request the AIAD placement, submit the SIEDS abstract, and register for the conferences. Between March and May 2027, I will attend the funded conference (dates in the table above), and I will complete the AIAD from 11–31 July 2027. I will buy each MITx certificate when that course opens, and all funds will be spent by 31 December 2027, before my graduation in May 2029.

5. The point of contact for this memorandum is CDT Dennis Pezan at (815) 592-6927 or dennis.pezan@westpoint.edu.

Encl
Applicant’s Curriculum Vitae

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